

OSPF Passive Interface Hardening & Optimization Lab

Network Engineering Lab Portfolio

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1 Overview

This technical report details the implementation and verification of OSPF passive interfaces using Cisco IOS routers in GNS3. The primary objective of this lab is to demonstrate control plane security hardening by suppressing unnecessary OSPF Hello multicast traffic on host-facing LAN subnets while maintaining complete routing table propagation across the backbone network.

2 Lab Prerequisites & Assumptions

- Single-Area OSPF (Area 0) configured across all three routers.
- Physical links initialized in an up/up operational state.
- Wireshark packet capture configured on the host access link between R1 (`eth1/0`) and PC1 (`e0`).

3 Network Topology & Addressing Schema

The lab topology features a three-router backbone (**R1**, **R3**, **R2**) with host subnets attached to perimeter routers **R1** and **R2**.

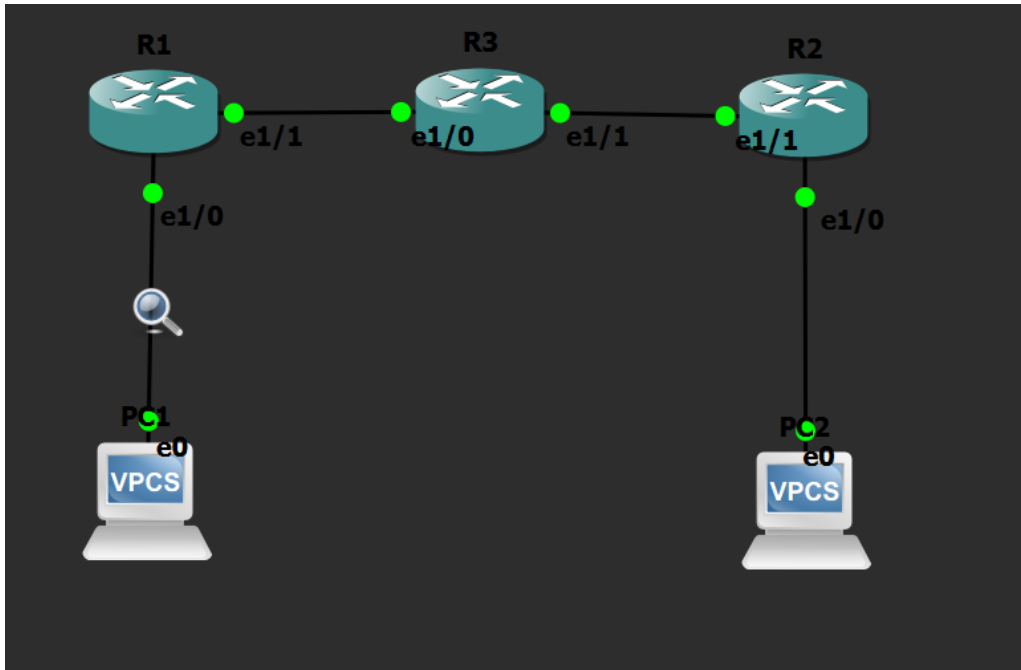


Figure 1: GNS3 Network Topology Diagram.

3.1 Addressing Chart

Device	Interface	IP Address	Subnet Mask	OSPF Area	Interface Status
R1	eth1/1	10.1.1.1	255.255.255.252	Area 0	Active Transit Link
R1	eth1/0	192.168.10.1	255.255.255.0	Area 0	Passive (PC1 Gateway)
R3	eth1/0	10.1.1.2	255.255.255.252	Area 0	Active Transit Link
R3	eth1/1	10.2.2.1	255.255.255.252	Area 0	Active Transit Link
R2	eth1/1	10.2.2.2	255.255.255.252	Area 0	Active Transit Link
R2	eth1/0	192.168.20.1	255.255.255.0	Area 0	Passive (PC2 Gateway)
PC1	e0	192.168.10.100	255.255.255.0	N/A	End Host
PC2	e0	192.168.20.100	255.255.255.0	N/A	End Host

Table 1: Lab Addressing and Interface Role Matrix.

4 Technical Deep Dive

4.1 Does a Passive Interface Change the Routing Table?

- **On the Local Router: No.** The local subnet attached to the passive interface remains installed in the Routing Information Base (RIB) as a directly connected route (C).
- **On Remote Routers: No.** Remote routers across the network continue to learn and install the subnet as a valid OSPF route (O).

The `passive-interface` command exclusively affects ****control plane traffic****. The router stops sending multicast Hello packets (224.0.0.5) and listening for neighbor requests on that port. However, the route information is still advertised to the rest of the network through active transit links via Link-State Advertisements (LSAs).

4.2 When is it Good Practice to Set an Interface as Passive?

1. **Host-Facing Access Links:** Workstations, servers, and IP phones do not run OSPF. Suppressing Hellos prevents unauthorized users from forming rogue adjacencies or attempting route injection attacks.
2. **Loopback Interfaces:** Software loopbacks used for Router IDs have no physical neighbors. Setting them to passive removes unnecessary control processing.
3. **External Boundaries:** Links facing external firewalls or ISP routers should be set to passive to prevent leaking internal topology details.

5 Step-by-Step Configuration Guide

5.1 Step 1: Baseline Router Configuration

Configure interface IP addresses across all three routers.

Router R1:

```
R1(config)# interface Ethernet1/1
R1(config-if)# ip address 10.1.1.1 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# interface Ethernet1/0
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# no shutdown
```

Router R3:

```
R3(config)# interface Ethernet1/0
R3(config-if)# ip address 10.1.1.2 255.255.255.252
R3(config-if)# no shutdown
R3(config-if)# interface Ethernet1/1
R3(config-if)# ip address 10.2.2.1 255.255.255.252
R3(config-if)# no shutdown
```

Router R2:

```
R2(config)# interface Ethernet1/1
R2(config-if)# ip address 10.2.2.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# interface Ethernet1/0
R2(config-if)# ip address 192.168.20.1 255.255.255.0
R2(config-if)# no shutdown
```

5.2 Step 2: Enterprise Security Hardening with "passive-interface default"

To implement security best practices, suppress OSPF globally by default on all interfaces, then explicitly un-suppress only the active transit links.

Router R1 Configuration:

```
R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# passive-interface default
R1(config-router)# no passive-interface Ethernet1/1
R1(config-router)# network 10.1.1.0 0.0.0.3 area 0
R1(config-router)# network 192.168.10.0 0.0.0.255 area 0
```

Router R3 Configuration:

```
R3(config)# router ospf 1
R3(config-router)# router-id 3.3.3.3
R3(config-router)# network 10.1.1.0 0.0.0.3 area 0
R3(config-router)# network 10.2.2.0 0.0.0.3 area 0
```

Router R2 Configuration:

```
R2(config)# router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# passive-interface default
R2(config-router)# no passive-interface Ethernet1/1
R2(config-router)# network 10.2.2.0 0.0.0.3 area 0
R2(config-router)# network 192.168.20.0 0.0.0.255 area 0
```

6 Verification & Wireshark Packet Analysis

6.1 Wireshark Capture Before Passive Interface Configuration

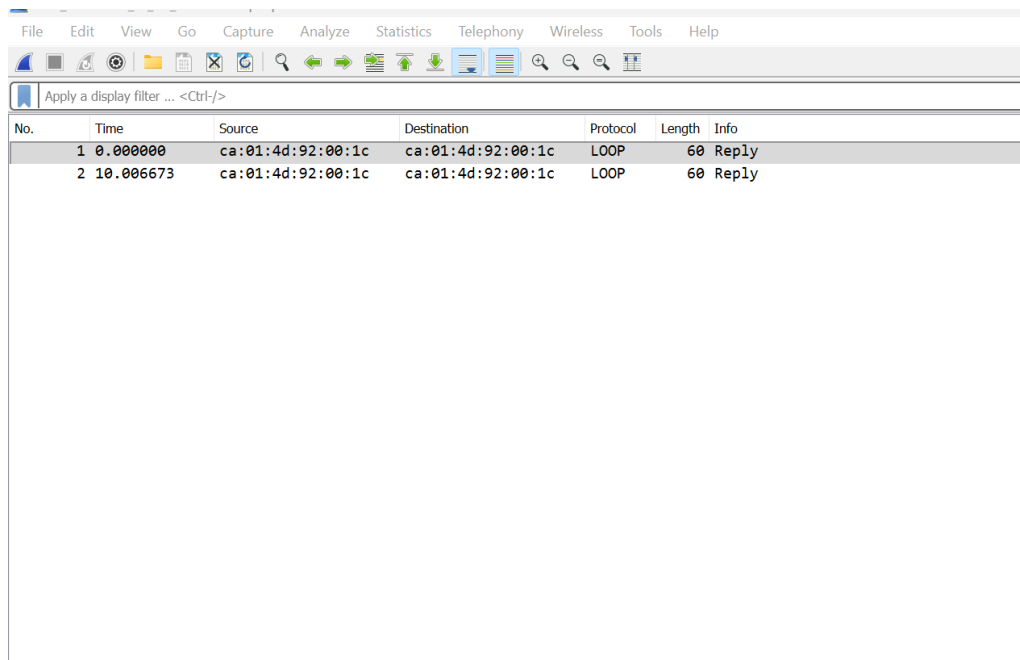
Before applying the passive interface command, a live packet capture on the R1-PC1 link (Ethernet1/0) recorded R1 actively sending multicast OSPF Hello packets to destination IP 224.0.0.5 every 10 seconds:

1	0.000000	192.168.10.1	224.0.0.5	OSPF	90	Hello Packet
2	4.592058	ca:01:4d:92:00:1c	ca:01:4d:92:00:1c	LOOP	60	Reply
3	9.338854	192.168.10.1	224.0.0.5	OSPF	90	Hello Packet
4	14.587764	ca:01:4d:92:00:1c	ca:01:4d:92:00:1c	LOOP	60	Reply
5	18.847701	192.168.10.1	224.0.0.5	OSPF	90	Hello Packet
6	24.590284	ca:01:4d:92:00:1c	ca:01:4d:92:00:1c	LOOP	60	Reply
7	27.845600	192.168.10.1	224.0.0.5	OSPF	90	Hello Packet
8	34.590486	ca:01:4d:92:00:1c	ca:01:4d:92:00:1c	LOOP	60	Reply
9	34.620644	ca:01:4d:92:00:1c	CDP/VTP/DTP/PAgP/U...	CDP	346	Device ID: R1 Port ID: Ethernet1/0
10	37.506666	192.168.10.1	224.0.0.5	OSPF	90	Hello Packet
11	44.589990	ca:01:4d:92:00:1c	ca:01:4d:92:00:1c	LOOP	60	Reply
12	47.347979	192.168.10.1	224.0.0.5	OSPF	90	Hello Packet
13	54.590073	ca:01:4d:92:00:1c	ca:01:4d:92:00:1c	LOOP	60	Reply
14	56.994726	192.168.10.1	224.0.0.5	OSPF	90	Hello Packet
15	64.588114	ca:01:4d:92:00:1c	ca:01:4d:92:00:1c	LOOP	60	Reply
16	66.478789	192.168.10.1	224.0.0.5	OSPF	90	Hello Packet
17	74.586877	ca:01:4d:92:00:1c	ca:01:4d:92:00:1c	LOOP	60	Reply
18	76.388200	192.168.10.1	224.0.0.5	OSPF	90	Hello Packet
19	84.587892	ca:01:4d:92:00:1c	ca:01:4d:92:00:1c	LOOP	60	Reply
20	86.187399	192.168.10.1	224.0.0.5	OSPF	90	Hello Packet
21	94.588078	ca:01:4d:92:00:1c	ca:01:4d:92:00:1c	LOOP	60	Reply
22	94.618271	ca:01:4d:92:00:1c	CDP/VTP/DTP/PAgP/U...	CDP	346	Device ID: R1 Port ID: Ethernet1/0
23	95.403730	192.168.10.1	224.0.0.5	OSPF	90	Hello Packet

Figure 2: Wireshark Capture Showing Active OSPF Hello Multicasts to 224.0.0.5.

6.2 Wireshark Capture After Passive Interface Configuration

Immediately after executing `passive-interface default` on R1, OSPF Hello packets ceased transmission on Ethernet1/0. The live capture shows that only non-routing loop frames remain:



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	ca:01:4d:92:00:1c	ca:01:4d:92:00:1c	LOOP	60	Reply
2	10.006673	ca:01:4d:92:00:1c	ca:01:4d:92:00:1c	LOOP	60	Reply

Figure 3: Wireshark Capture Verifying Complete OSPF Hello Suppression.

6.3 Control Plane CLI Verification

Running `show ip ospf interface Ethernet1/0` on R1 explicitly confirms that Hello generation is disabled:

Listing 1: R1 OSPF Interface State Output

```
R1#show ip ospf interface Ethernet1/0
Ethernet1/0 is up, line protocol is up
  Internet Address 192.168.10.1/24, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 192.168.10.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
  No Hellos (Passive interface)
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 0
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
```

6.4 Data Plane Connectivity Test

To verify that passive subnets remain reachable, an ICMP ping was executed from PC1 (192.168.10.100) to PC2 (192.168.20.100) across the backbone:

Listing 2: PC1 to PC2 End-to-End Ping Output

```
PC1> ping 192.168.20.100
84 bytes from 192.168.20.100 icmp_seq=1 ttl=62 time=28.104 ms
84 bytes from 192.168.20.100 icmp_seq=2 ttl=62 time=26.432 ms
```

7 Conclusion

This lab validates that configuring `passive-interface default` secures host-facing subnets by eliminating unnecessary OSPF Hello traffic to 224.0.0.5. Control plane verification and packet captures prove that interface passivation optimizes network performance and hardens security without impacting routing table propagation or data plane reachability.